

Modeling time-varying survival in North Pacific humpback whales

More exciting title coming soon!

Navigation

Abstract	2
Introduction	2
Materials and Methods	2
Data Sources	2
Stock Structure	2
International Whaling Commission	2
Abundance data	3
Removals	4
Mixing data	4
Environmental covariate data	4
Population Model	5
Mortality Attribution	6
Model Variations	7
Environmental variable selection	8
Observation Model	8
Estimation	9
Results	10
Discussion	10
Tables	10
Figures	10

Abstract

Introduction

Materials and Methods

We used data on abundance, stock structure, and removals, combined with environmental covariate data derived from an ocean model, to parameterize population dynamics models for North Pacific humpback whales. We designed four separate models which all had the same general structure, but varied in their specification of annual survival. Our general objectives were a) to assess whether and how humpback survival has varied since 2000; and b) to determine whether the inclusion of environmental information improves our estimation of survival. In the following, we describe in detail the data, models, diagnostics, and derived quantities of interest. Table ZZ reports model similarities and differences.

Data Sources

Stock Structure

The population model is structured with multiple individual feeding and breeding grounds, or zones. This stock structure for the North Pacific humpback population was developed through IWC assessment process, and includes distinct breeding grounds in mainland Mexico, Mexican offshore islands, Central America, Asia, and Hawai'i, and feeding zones in Oregon/California, southern British Columbia/Washington, Southeast Alaska and northern British Columbia, the Gulf of Alaska, the eastern Aleutian islands/Bering Sea, and Russia/western Aleutians (Figure XX). Therefore, the model requires data on abundance both within and across these zones, as well as information on the mixing of whales between zones (i.e., the proportional allocation of humpbacks from breeding zones to feeding zones, and from feeding zones to breeding zones), and data on direct, human-caused mortality due to bycatch, entanglement, and ship strikes.

International Whaling Commission

brief description of assessment history and objectives

Abundance data

During the IWC assessment process, relative and absolute abundance data were collated from multiple surveys to inform the population model (Table SX). We gathered time series data on total (North Pacific-wide) abundance, as well as the abundance within individual feeding and breeding grounds. One source of abundance data was the SPLASH (Status of Populations, Levels of Abundance, and Status of Humpbacks) effort from 2004-2006. The SPLASH project used mark-recapture photo-identification sampling in both breeding and feeding zones of the North Pacific. @wade2021 estimated absolute abundance from a multi-strata model based on this mark-recapture sampling, and we adopt those estimates for most zones, with the Oregon/California feeding ground and Mexico breeding ground excepted. For the Mexico breeding zone, we used data from @martinez2017, who used a dataset moderately expanded from the SPLASH data (i.e., more years) to compute similar mark-recapture estimates to @wade2021. For the Oregon/California feeding zone, we use abundance estimates from @calambokidis2024, who used a closed-population Chao Mth model to estimate abundance, while accounting for heterogeneity in capture probabilities. @curtis2022 made an additional estimate for the Central America breeding zone for 2019-2021, using a one-dimensional spatial capture-recapture model. @baker1987 made an additional estimate for Hawai'i for 1980-1983, @calambokidis2017 provide an estimate for Hawai'i for 1991-1993, and @calambokidis1997 provide an estimate for the Asia breeding zone for 1991-1993.

For the Alaskan feeding zones, including the Bering Sea, Aleutian Islands, and Gulf of Alaska, we use data from the IWC-POWER (Pacific Ocean Whale and Ecosystem Research Programme) surveys. Because the survey strata from the POWER surveys do not exactly match the spatial zones in the present analysis, @matsuoka2023 prorated (translated) the POWER estimates to our model zones by counting the number of individual whales seen in each model zone, based on maps from the POWER cruise reports. For the Gulf of Alaska, we also use estimates from @zerbini2006 which derive from shipboard line-transect methods for the years 1987 and 2001-2003.

For estimates of total population size across zones, we use data from @cheeseman2024, who estimated humpback abundance for each year from 2001-2021 by utilizing an extensive (and growing) catalog of research-based and citizen science photo-identification data collected through Happywhale.com, including data from the original SPLASH project. @cheeseman2024 applied a Chapman-Petersen estimator for abundance, and used multiple methods to address potential biases in the data, including using bias correction to account for geographically biased sampling during the years of the SPLASH project.

Additional data provide estimates of relative, rather than absolute abundance. These data were incorporated as relative indices of abundance usually because they sampled only a subset of the spatial area associated with a particular zone. These include time series of relative abundance estimates for Hawai'i from @cheeseman2024 and @moble2001, for Mexico from @cheeseman2024, for the Gulf of Alaska from @moran2023 and @arimitsu2022, for Southeast Alaska from @gabriele2022, and for the Bering Sea from @friday2012.

Importantly, every abundance datum is associated with a coefficient of variation, allowing to the extent possible for the uncertainty from the original studies to be appropriately incorporated into the present analysis.

Removals

Because humpbacks have been subject to human-caused mortality, we gathered estimates of removals of humpback whales due to directed hunting, bycatch, and ship strikes (Table SX). The IWC maintains data on commercial catches (which had largely ceased by the time our model begins in 1970). An analysis by @palka2024 provides the estimates of bycatches and ship strikes, using data from fishery observers, humpback stranding records, and previous IWC reports.

Mixing data

Estimates of mixing between breeding and feeding zones come from two data sources that each calculate both forward (feeding to breeding) and reverse (breeding to feeding) mixing matrices. The first dataset is from @steel2024, who use mixed stock analyses of maternally inherited mtDNA haplotypes combined with biparentally inherited nuclear microsatellite markers to estimate forward apportionment from feeding areas to breeding areas for each breeding-feeding hypothesis. In the same study, @steel2024 estimated the reversed apportionment from breeding areas to feeding areas, based only on mtDNA haplotypes (Table SX).

The second dataset for mixing comes from @wade2023, who used a multi-state mark recapture model to estimate the forward and reverse mixing matrices based on the SPLASH photo-identification dataset (Table SX).

Environmental covariate data

We collected data on environmental covariates hypothesized to affect humpback survival in the North Pacific [@arimitsu2022;@gabriele2022]. We extracted covariate data from the GLORYSv12 (Global Ocean Reanalysis and Simulation) model from Mercator-Ocean [@leloulouche2018]. The GLORYS suite of models provides both physical (0.083-degree spatial resolution) and biogeochemical (0.25-degree resolution) variables globally from 1993-present. Candidate environmental variables included sea surface temperature, mixed layer depth, chlorophyll, nitrate, and net primary production during the boreal summer (June to August). Using the spatial boundaries of the breeding and feeding grounds (Figure SX), we calculated zone-specific annual values for each environmental variable. Specifically, we calculated anomaly time series, representing deviations from a 1993-2013 variable-specific climatology. When incorporating them into the population model, each time series was normalized (z-scored) by subtracting the mean and dividing by the standard deviation.

Population Model

The population model for North Pacific humpback whales is an age-structured, sex-aggregated population dynamics model with multiple breeding stocks. Within the model, “herds” of whales have fidelity to a particular breeding and feeding ground combination. Each herd begins a model year on the breeding grounds, where direct removals on the breeding grounds are removed. Subsequently, on the feeding grounds, herds are subject to additional direct removals, as well as natural mortality (i.e., natural mortality is assumed to occur due to conditions on the feeding grounds, not the breeding grounds). Finally, at the end of the model year is when fecundity—and therefore next year’s starting abundance—is calculated.

The general form of the population dynamics is:

$$N_{y+1,a}^{b,f} = \begin{cases} B_{y+1}^{b,f} & \text{if } a = 0 \\ \tilde{N}_{y,a-1}^{b,f} S_{y,a-1}^{b,f} & \text{if } 0 < a < x \\ \tilde{N}_{y,x-1}^{b,f} S_{y,x-1}^{b,f} + \tilde{N}_{y,x}^{b,f} S_{y,x}^{b,f} & \text{if } a = x \end{cases} \quad (1)$$

where $N_{y,a}^{b,f}$ is the number at the start of year y of age a animals in the herd defined by breeding ground b and feeding ground f , $S_{y,a}^{b,f}$ is their survival rate in year y , and $\tilde{N}_{y,a}^{b,f}$ is the number after all removals during year y :

$$\tilde{N}_{y,a}^{b,f} = \tilde{N}_{y,a}^{b,f} \left(\tilde{N}_{y,0+}^f - C_y^f \right) / \tilde{N}_{y,0+}^f \quad (2)$$

where C_y^f is removals from feeding ground f in year y , and $\tilde{N}_{y,0+}^f = \sum_b \sum_a \tilde{N}_{y,a}^{b,f}$ is the total number of animals on feeding ground f across herds and ages. The number of animals in each herd after breeding ground removals, but *before* feeding ground removals, is given by:

$$\tilde{N}_{y,a}^{b,f} = N_{y,a}^{b,f} (N_{y,0+}^b - C_y^b) / N_{y,0+}^b \quad (3)$$

where C_y^b is removals from breeding ground b in year y , and $N_{y,0+}^b$ is the total abundance of breeding stock b at the start of year y , $N_{y,0+}^b = \sum_f \sum_a N_{y,a}^{b,f}$.

In the base version of the model, fecundity depends on the density of whales on the feeding grounds. The number of births for each herd, $B_y^{b,f}$, is determined by

$$B_y^{b,f} = 0.5 f_y^f N_y^{b,f} \quad (4)$$

where f_y^f is the fecundity for animals in feeding ground f . Fecundity varies between a minimum and maximum value,

$$f_y^f = f_0 - (f_{\max} - f_0) \left(1 - \frac{N_{y,1+}^f}{K_{1+}^f} \right) \quad (5)$$

where f_0 is fecundity at carrying capacity, $f_{\max}$ is fecundity at zero population size, K_{1+}^f is the age 1+ carrying capacity for feeding ground f , and $N_{y,1+}^f = \sum_b \sum_{a>0} \widetilde{N}_{y,a}^{b,f}$. f_0 is determined from Equation 1 under unfished equilibrium conditions, and $f_{\max}$ is calculated from a reparameterized version of the Lotka equation [Zerbini2010]:

$$f_{\max} = \frac{(1 + r_{\max})^{L+1} - S_A(1 + r_{\max})^L}{S_A^L S_C} \quad (6)$$

where $r_{\max}$ is the maximum rate of increase and L is age-at-maturity. S_C and S_A are base-case survival for age 0 and age 1+ humpbacks, respectively.

Annual survival is a key quantity of interest in this study. In the base version of the model, survival is given by:

$$S_{y,a}^{b,f} = \begin{cases} \frac{1}{1 + \exp(\varphi_C + \varepsilon_y^f)} & (\text{calves, } a = 0) \\ \frac{1}{1 + \exp(\varphi_A + \varepsilon_y^f)} & (\text{adults, } a > 0) \end{cases} \quad (7)$$

$$\varphi_C = \ln\left(\frac{1}{S_C} - 1\right); \quad \varphi_A = \ln\left(\frac{1}{S_A} - 1\right) \quad (8)$$

where ε_y^f is the survival deviation for feeding ground f in year y .

Initial conditions for each herd are determined by the total abundance at the start of the first model year, assuming an equilibrium population structure relative to S_A and S_C .

Mortality Attribution

The population model tracks annual survival, and therefore also tracks mortality. Additionally, we are interested in estimating the contribution of natural mortality to total mortality, relative to the effect of direct removals. Specifically, we want to track “excess” realized mortality due to survival deviating below its baseline value S_A . To estimate this excess mortality, we calculate, for each year, the difference between realized mortality of age 1+ whales (including the effect of any annual deviations) and a counterfactual scenario where survival is fixed at its baseline value S_A :

$$M_y^{b,f} = (S_{y,1+}^{b,f} - S_A) N_{y,1+}^{b,f} \quad (9)$$

We can then compare these $M_y^{b,f}$ values to C_y^b and C_y^f , and can track $\sum_b \sum_f M_y^{b,f}$ cumulatively across time to obtain an estimate of total excess mortality across the modeled period.

Model Variations

A main objective of our analysis was to evaluate evidence for alternative specifications of annual survival, as well as to determine whether environmental data can help improve our estimation of survival. Therefore, we defined four models, representing different hypotheses for how environmental variability affects humpback populations (with model shorthand in **bold**).

The base version of the model calculates survival according to Equation 7, with annual survival ε_y^f specified as random normal deviates.

$$\varepsilon_y^f \sim \mathcal{N}(0, \sigma_\varepsilon^2) \quad (10)$$

We call this the random-survival (**rS**) model.

In the environment-survival (**envS**) model, we introduce environmental influences on survival by representing ε_y^f as a function of environmental covariates:

$$\omega^f \Upsilon_y^f + \epsilon_y^f \quad (11)$$

where ω^f is a vector of coefficients for the environmental covariates derived for feeding ground f , Υ_y^f is a vector of environmental covariate data for feeding ground f in year y , and together $\omega^f \Upsilon_y^f$ is our environmental index of survival. $\epsilon_y^f \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is the remaining (unexplained) variation in survival. Realized survival is then calculated through the logistic function in Equation 7, meaning that in the **rS** model, larger values of the environmental index correspond to an increasingly negative effect on survival.

Alternatively, annual survival plausibly could be due to density-dependence only, with no other deviations, environmentally-driven or otherwise. In the density-dependence only model (**ddOnly**), deviations in annual survival are driven by Beverton-Holt density-dependence with respect to an estimated carrying capacity K^f for each humpback feeding ground, and we replace Equation 7 with,

$$S_{y,a}^{b,f} = \begin{cases} \frac{S_C}{1 + \alpha_S(\frac{N_{y,1+}^f}{K_{1+}^f})} & (\text{calves, } a = 0) \\ \frac{S_A}{1 + \alpha_S(\frac{N_{y,1+}^f}{K_{1+}^f})} & (\text{adults, } a = 1+) \end{cases} \quad (12)$$

where α_S represents the strength of density-dependence in survival. In this model (and the **envK** model described below), we estimate a similar density-dependence for fecundity, replacing Equation 6 with the following:

$$f_y^f = \frac{f_{max}}{1 + \beta_S(\frac{N_{y,1+}^f}{K_{1+}^f})} \quad (13)$$

where β_S represents the strength of density-dependence in fecundity.

Finally, it is possible that environmental variation could engender changes in carrying capacity rather than having a direct effect on survival. Our environment-K (**envK**) model is based on the formulation @stewart2023 used for understanding changes in gray whale survival. In the **envK** model, we introduce time-varying carrying capacity for each feeding ground:

$$K_y^f = \bar{K}^f \exp(\omega^f \Upsilon_y^f + \epsilon_y^f) \quad (14)$$

where $\bar{K}^f$ is the estimate of long term mean carrying capacity for feeding ground f , and the environmental index is defined as in Equation 11. The **envK** model then uses Equation 12 and Equation 13 to calculate survival and fecundity, respectively.

Environmental variable selection

We used a variable selection process to select which combination of the five environmental covariates resulted in the most parsimonious model, balancing model fit with model complexity (number of parameters). We fit the model (see below) for every possible combination of the five environmental covariates ($n=31$), and selected the set of covariates that minimized the value $2P - 2\Gamma$, where P is the number of environmental parameters (one per feeding ground per environmental covariate) and Γ is the summed log likelihood of the survival deviates given σ_ϵ^2 . To retain comparability among models, we performed this selection exercise with the **rS** model, then used the selected variables to fit the other three models.

Observation Model

Generically, the observations of absolute abundance described above are linked to the population model as:

$$n_y \sim \mathcal{N}(N_y, \sigma_{n_y}) \quad (15)$$

where n_y is an estimate of abundance (i.e., observation) in year y with standard deviation σ_{n_y} (Table SX). Each estimate is normally distributed with mean N_y the model-estimated abundance. We use this generic form for abundance estimates linked to different age groups (age-0+ or age-1+), and for observations collected on breeding or feeding grounds (i.e., N_y in Equation 15 can represent total abundance N_y , breeding ground abundance N_y^b , or feeding ground abundance $\tilde{N}_y^f$).

The observations of relative abundance are represented in a similar way,

$$n_y \sim \mathcal{N}(q_x N_y, \sigma_{n_y}) \quad (16)$$

where q_x is the constant of proportionality (i.e., survey bias) associated with the relative index of abundance that scales each survey x from relative to absolute. Maximum likelihood values of q_x are calculated analytically in the model.

The proportion of breeding ground b animals that migrate to feeding ground f in year y is:

$$\rho_y^{b,f} = N_{y,1+}^{b,f} / N_{y,1+}^b \quad (17)$$

with a similar calculation for the reverse migration from feeding to breeding grounds. These proportions together form the mixing estimates—both forward and reverse apportionment of animals between breeding and feeding grounds. The mixing estimates are assumed to be Dirichlet-distributed based on the observations from both mixing datasets (genetics and mark-recapture).

need to finish this and explain the Dirichlet calculation and sample size

Note that in the model, removals are fixed (i.e., we use the removals data in the model, but they do not enter the observation model). This decision was made to improve identifiability and reduce the total number of estimated parameters.

Estimation

The key estimated parameters in the model include maximum growth rate r_{max} , carrying capacity for each breeding ground K^b , the relative depletion of each stock at the beginning of the modeled period $N_{y=0}^b / K^b$, the mixing proportions, the annual survival deviations ε_y^f (or carrying capacity deviations in the case of the **envK** model), the Beverton-Holt density-dependence parameters for the **ddOnly** and **envK** models, and the coefficients of the environmental index ω^f . In addition, the sampling coefficients of variation reported in the data appear to be very low for three of the observed time series of abundance, so we introduce additional variance parameters for estimates of absolute abundance for the SBC+WA and OR+CA feeding grounds and the relative index of abundance for the GOA feeding ground. Vague or flat priors were used for all parameters.

maybe more about priors.

The model is written in *RTMB* [rtmb] but passed to Stan for Bayesian MCMC sampling using *rstan* and *tmbstan* [rstan;tmbstan]. We ran each of the four models (**rS**, **ddOnly**, **envS**, and **envK**) with four chains of 10,000 iterations each. We used standard diagnostics to assess model convergence. Specifically, we visually inspected trace plots for adequate chain mixing, ensured that the model did not have divergent transitions, that each estimated parameter had an effective sample size >1500 for all fitted parameters, and that the Gelman-Rubin convergence diagnostic (R-hat) was less than 1.01. Model fit was assessed based on the fits to the various indices of abundance.

maybe need more about assessing model fit

Model	Shorthand	Survival
Random survival	rS	Random effect
Environment-survival	envS	Survival indexed by environmental covariates
Density-dependence only	ddOnly	Beverton-Holt, with no environmental covariates
Environmental-K	envK	Carrying capacity indexed by environmental covariates, with Beverton-Holt

Results

Discussion

Tables

```
# summary of models
#|label: tbl-models
#|tbl-cap: "Summary of the population models used to assess time varying humpback survival."

df <- data.frame(
  Model=c("Random survival","Environment-survival","Density-dependence only","Environmental-K"),
  Shorthand=c("rS","envS","ddOnly","envK"),
  Survival=c("Random effect","Survival indexed by environmental covariates","Beverton-Holt, with no environmental covariates","Carrying capacity indexed by environmental covariates, with Beverton-Holt")
)
kable(df, escape = FALSE, format = "latex") |>
  kable_styling(bootstrap_options = "striped")
```

Figures